

On-Board Real-Time AI Compensation System for Nonlinear Drift in Electrochemical Seismometers

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Abstract

Electrochemical seismometers (MET) are gaining importance in geophysical exploration due to their high sensitivity, wide bandwidth, and strong tilt/impact resistance, and are considered a promising next-generation seismic detection solution. However, complex coupling among mechanical, fluid, and electrochemical processes induces significant magnitude-dependent nonlinearity, causing frequency response drift and limiting dynamic range, ultimately degrading seismic data quality and event detection reliability. Traditional force feedback methods risk closed-loop oscillation at large magnitudes, while data-driven deep learning methods, despite high accuracy in nonlinear compensation, struggle to meet real-time requirements on resource-constrained microcontrollers due to intensive matrix operations. To address real-time compensation and feedback-stability limitations, this paper proposes Wiener-KAN, a feedforward compensation system for on-board real-time execution. The system adopts the Wiener-model decomposition into linear dynamics and static nonlinearity: in the linear part, the Wiener linear layer is initialized with the relative frequency response between the target and ideal systems, injecting a physical prior and enabling rapid frequency feature extraction; in the nonlinear part, a sign- and symmetry-constrained Kolmogorov-Arnold network (KAN) with trainable spline activation functions is introduced and precomputed into lookup tables, replacing spline-function evaluation with low-latency table lookup during inference. Furthermore, this paper constructs and

releases a three-dimensional frequency response dataset capturing amplitude-frequency coupling effects, and designs an amplitude-frequency loss (AFMAE) targeting frequency characteristic drift to improve frequency-domain fitting accuracy. Experimental results show that with input magnitude increased by over $10\times$, Wiener-KAN reduces frequency drift from 34.50 Hz to 2.34 Hz, 100 Hz sensitivity drift from $84.56\text{ V}\cdot\text{s}/\text{m}$ to $9.09\text{ V}\cdot\text{s}/\text{m}$, and linearity error from 2.004% to 0.511%. For real-time feasibility, the precomputed KAN lookup table converts complex floating-point operations into ultra-low-latency memory addressing, achieving a throughput of 6986.7 points/s and a minimal RAM footprint of 5.6 KB on the STM32F405 microcontroller, providing an effective solution for real-time MET nonlinear compensation.

Keywords: electrochemical seismometer, Kolmogorov-Arnold network, nonlinearity, frequency response, loss function, on-board real-time compensation, compensation system

1 Introduction

Electrochemical seismometers (MET) are emerging as promising candidates for next-generation seismic sensors in underground resource exploration and earthquake monitoring, owing to their high sensitivity, wide bandwidth, and strong tilt/impact resistance [1]. As global seismic networks progressively adopt real-time data acquisition with wide bandwidth and high dynamic range [2], core sensors face stricter requirements on linearity and response fidelity. However, the complex coupling between electrochemical reactions and fluid mechanics introduces significant magnitude-dependent nonlinearity into the MET frequency response, including sensitivity drift and natural frequency drift. Such drift distorts vibration waveform detection, limits dynamic range, and degrades seismic survey data quality and event identification reliability.

Prior research has attempted to improve linearity through structural design optimization. Xu et al. [3] proposed a MEMS electrochemical seismometer using an anodic bonding process, enhancing sensitivity and device consistency via a silicon-glass-silicon three-layer integrated electrode structure. Sun et al. [4] developed a single-silicon-chip four-microelectrode wide-bandwidth electrochemical seismometer, achieving an operating bandwidth of 0.7–100 Hz and a peak sensitivity of $6221\text{ V}\cdot\text{s}/\text{m}$ at 3 Hz. Such structural optimization offers limited improvement in linearity; systems typically maintain quasi-linear operation only below $1.0\text{ mm/s}@20\text{ Hz}$ (equivalent acceleration of 0.13 m/s^2) [3], or below $0.7\text{ mm/s}@20\text{ Hz}$ (equivalent acceleration of 0.09 m/s^2) [4].

Force feedback is an effective means of nonlinear suppression. For example, Li et al. [5] and Sun et al. [6] reported reductions in MET sensitivity drift or input-output nonlinear error using force feedback. However, force feedback requires additional mechanical feedback units and is limited by reduced nonlinear phase margin and closed-loop oscillation risk at large magnitudes. Sensor calibration methods based on predictive models [7] demonstrate that software-based feedforward compensation improves sensor output quality while maintaining hardware simplicity.

This feedforward direction is directly relevant to MET sensors. The MET nonlinearity is mainly expressed as a magnitude-dependent equivalent frequency response, while the

acquisition chain can remain fixed. A compensator can therefore correct the measured time-domain signal without modifying the electrochemical chamber, mechanical vibration pickup unit, or readout circuit. Compared with feedback force applied in real time to the mechanical structure, software feedforward compensation turns the stability issue into an input-output mapping problem on measured data, making it compatible with edge microcontrollers, digital acquisition chains, and batch calibration workflows.

Recent studies show that deep neural networks can achieve high sensor compensation accuracy when trained with sufficient time- and frequency-domain test data [8]. This paper performs steady-state testing on an electrochemical seismometer covering the full bandwidth from 10 Hz to 128 Hz, and extends the input upper limit from 0.46 m/s^2 in [6] to 6.0 m/s^2 , an increase of over $10 \times$. Based on this, we construct and release a three-dimensional frequency response dataset that includes magnitude, frequency, and input-output time series while explicitly recording amplitude-frequency coupling effects.

The dataset is more than a calibration curve at a single frequency or magnitude. The frequency dimension records response changes at the low-frequency end, resonance region, and high-frequency end; the magnitude dimension records the transition from the small-signal quasi-linear regime to the large-signal nonlinear regime; and the time-domain sequences preserve phase, amplitude, and local waveform morphology. These three forms of information jointly support the evaluation of frequency drift, sensitivity drift, and linearity error in this work.

After relatively complete test data are obtained, the core challenge becomes the design of an efficient nonlinear compensation algorithm. Conventional signal-processing identification methods [9–12] usually have difficulty determining model parameters accurately in strongly coupled electrochemical-fluid systems. Data-driven time-series models can improve compensation accuracy [13–15], but their parameter counts, matrix multiply-accumulate operations, and memory costs are difficult to reconcile with real-time processing on resource-constrained microcontrollers. Purely data-driven models also usually ignore sensor-specific frequency response characteristics, while networks that directly inject physical equations [16] are hard to adapt to MET sensors that are difficult to formulate accurately as PDE systems.

The resulting algorithmic target has three implications. First, the compensator must exploit the fact that MET remains approximately locally linear at a fixed magnitude, so that frequency response information enters the network explicitly. Second, the nonlinear mapping must have sufficient magnitude-dependent expressive capacity to handle residual coupling among different local dynamics. Third, the trained model must be converted into a low-latency and low-memory on-board implementation, enabling the compensation process to run point by point along a continuous sampling stream.

To address the limitations of computational constraints and missing physical priors, this paper proposes Wiener-KAN as an on-board real-time AI compensation system combining a Wiener linear layer with a Kolmogorov-Arnold network. The system adopts the classical Wiener model concept of separating linear dynamics from static nonlinearity [17]: in the linear part, measured local frequency response slices are mapped to a Wiener linear layer initialized by magnitude conditions; in the nonlinear part, a sign- and symmetry-constrained Kolmogorov-Arnold network (KAN) with trainable spline activation functions is introduced [18–21]. After training, the one-dimensional spline mappings in KAN are precomputed

into lookup tables, transforming dense floating-point operations into low-latency memory addressing and table lookup during on-board inference.

Accordingly, the contribution of this work is organized around an on-board compensation system and its data, model, loss, and deployment components. The three-dimensional frequency response dataset supplies measured evidence for amplitude-frequency coupling; the Wiener linear layer carries the local frequency response prior; the sign- and symmetry-conditioned KAN maps the remaining magnitude-dependent nonlinearity with trainable spline functions; AFMAE aligns the training objective with frequency-response accuracy; and the precomputed LUT converts the trained mapping into an embedded implementation. These components are evaluated under the same physical metrics and on-board protocol.

2 Background and Principles

2.1 Structure and Operating Principle of MET Electrochemical Seismometers

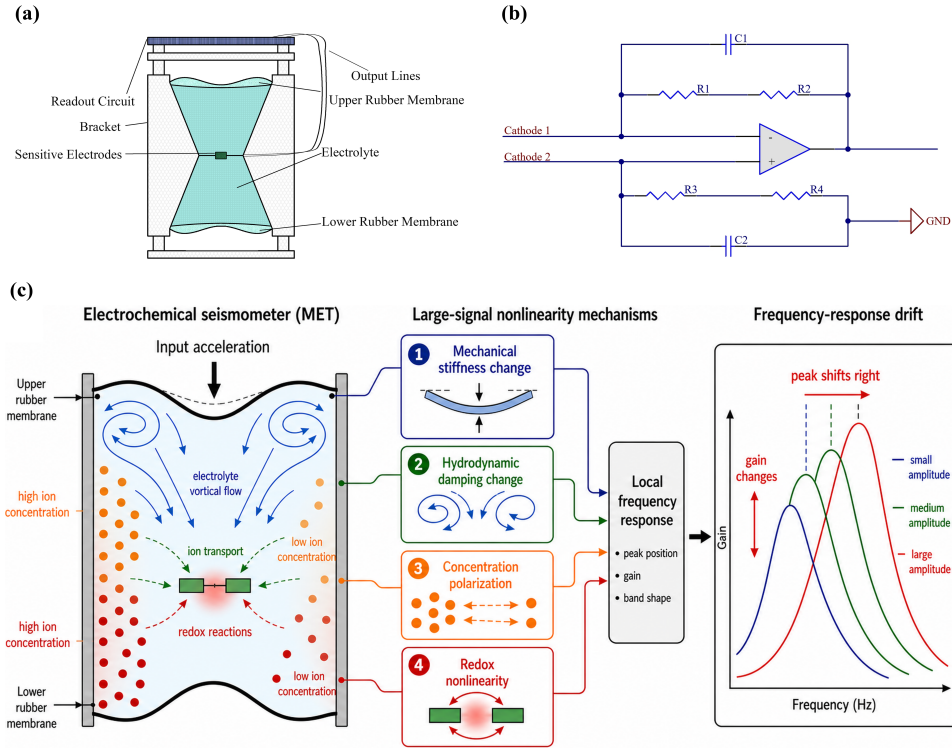


Fig. 1 MET sensing object, readout circuit, and large-signal nonlinear mechanism. (a) Schematic diagram of the MET structure. (b) Readout circuit of the MET. (c) Mechanism schematic of large-signal nonlinearity in MET, where mechanical geometry, fluid inertia, ion-transport coupling, and electrode kinetics jointly produce amplitude-dependent local dynamics.

The electrochemical seismometer (MET) is a high-sensitivity sensor that detects seismic vibrations based on molecular electron transfer. Its core components include an electrochemical chamber, sensing electrodes, electrolyte, and a mechanical vibration pickup unit (rubber membrane and mass body), as shown in Fig. 1(a). The readout circuit uses an AD706 operational amplifier to convert the weak differential current generated by the MET electrodes into a voltage signal, as shown in Fig. 1(b). When a seismic wave acts on the system, the mechanical structure deforms, driving electrolyte flow and altering the ion concentration distribution in the electrode region, thereby modulating the electrode reaction current density.

Under small-signal conditions, the MET frequency response can be obtained by cascading the mechanical vibration pickup subsystem and the electrochemical conversion subsystem, and the corresponding mechanical and electrochemical small-signal submodels have been established in prior MET studies [22–26]. The mechanical subsystem converts external vibration into volumetric flow rate of the electrolyte in the flow channel, and the electrochemical subsystem converts the flow-induced ion concentration perturbation into electrode current. The ideal small-signal response can therefore be described by a mechanical second-order high-pass stage and an electrochemical first-order low-pass stage, where the mechanical parameters govern the low-frequency end and resonance region, while ion diffusion near the electrodes controls the high-frequency end. The full derivation is given in Supplementary Note 1.

This small-signal model provides a common reference for the metrics and model design used later in this paper. The natural frequency summarizes the dynamic shift at the low-frequency end and resonance region, the sensitivity describes passband gain variation, and the damping ratio reflects the response-curve width and peak shape. Although the complete multiphysics process under large-signal operation is difficult to express as an exact closed-form model, these equivalent parameters compress the coupled response into measurable and comparable frequency-response features.

As input magnitude increases, the small-signal linearization condition gradually deviates from the actual operating state. Mechanical geometric nonlinearity, fluid inertial nonlinearity, multiplicative ion-transport coupling, and the exponential nonlinearity of electrode reaction kinetics jointly alter the local frequency response, appearing as magnitude-dependent shifts in equivalent natural frequency, sensitivity, and damping ratio, as shown in Fig. 1(c). The formula expansion of this amplitude-frequency coupling mechanism is provided in Supplementary Note 1.

This paper therefore treats the local frequency response at a fixed magnitude as an intermediate representation linking the physical mechanism and the learning model. This representation avoids directly solving the complete electrochemical-fluid nonlinear equations while preserving the dominant dynamic changes observed under different input magnitudes. The subsequent Wiener linear layer uses this local linear characteristic to organize relative transfer functions under multiple magnitude conditions into parallel linear kernels.

3 Experimental Design and Dataset

Precise evaluation and compensation of the amplitude-dependent nonlinear drift of MET sensors require comprehensive characterization of their dynamic response. Conventional single-dimensional frequency sweeps [27] cannot reveal system dynamics induced by input magnitude variations. This paper therefore performs two-dimensional sweeps in the joint

frequency–magnitude space and preserves complete input-output time series to record waveform distortion, phase shift, and local dynamic details simultaneously.

This experimental design places nonlinear characterization, training-sample construction, and result evaluation within the same frequency–magnitude matrix. Each operating condition provides both the measured uncompensated MET response and the ideal linear reference output under the same input condition. Thus, the training stage learns a compensation mapping from the measured output to the target linear output, while the evaluation stage returns to the same physical matrix to examine whether frequency response drift is suppressed.

3.1 Experimental Setup

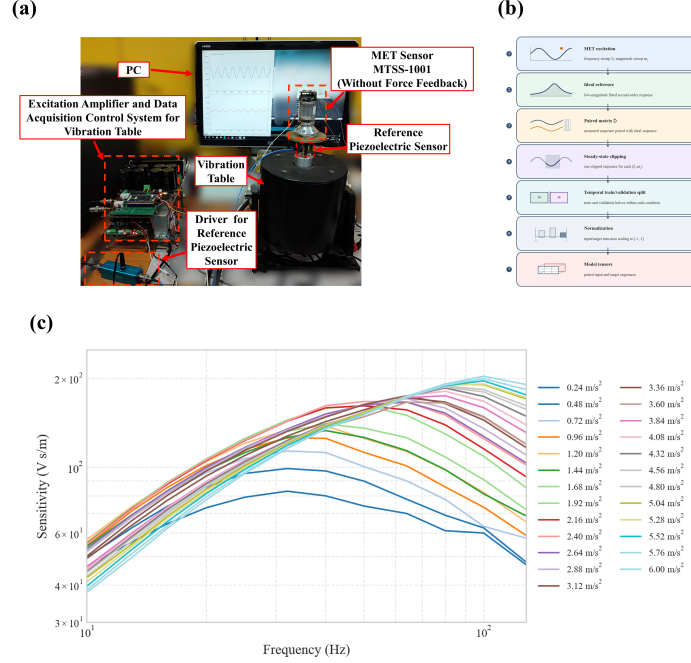


Fig. 2 Experimental setup, dataset construction workflow, and nonlinear response characterization. (a) Photograph of the experimental setup. (b) Illustrated dataset construction and preprocessing workflow. Controlled vibration-table sweeps define the frequency–magnitude operating grid, measured MET outputs are paired with the ideal linear reference, each clipped steady-state sequence is divided into temporal training/validation halves, and the input/target sequences are min-max normalized to $[-1, 1]$ before tensor packaging. (c) Measured nonlinear frequency responses of the uncompensated MET under the frequency–magnitude sweep.

As shown in Fig. 2(a), the vibration table generates steady-state sinusoidal signals according to the frequency–magnitude matrix. The MET sensor converts these signals into electrical outputs, which are recorded by the data acquisition system. The device under test is a MTSS-1001 electrochemical chamber without force feedback [28]. The test matrix covers 12 frequency sequences and 25 magnitude sequences, with a frequency range of 10–128 Hz, a

magnitude range of 0.24–6.0 m/s², and a sampling frequency of 2000 Hz. All frequency–magnitude conditions share identical excitation, acquisition, and reference channel settings; only the target frequency and magnitude are changed during recording. Detailed experimental parameters are listed in Supplementary Note 3.

Keeping the excitation, acquisition, and reference channel settings consistent during acquisition reduces extra differences introduced by changes in the test chain. The variation of the frequency response curves with magnitude is therefore mainly attributed to the amplitude-dependent dynamics of the MET itself, while the external acquisition conditions remain consistent across operating points. This setting also allows the low-amplitude reference system, uncompensated response, and compensated output to be compared in the same coordinates.

3.2 Dataset

The dataset obtained from the above experiments contains 300 frequency–magnitude operating conditions, with 8000 steady-state sampling points saved per condition. It uses paired time-domain sequences as basic samples, preserving both amplitude-frequency feature points and the pointwise waveform, phase, and local dynamic information required for end-to-end compensator training. The dataset can be represented as an $F \times M$ matrix $\mathcal{D}$:

$$\mathcal{D} = \left\{ \left(\tilde{\mathbf{Y}}_{i,j}, \overset{\circ}{\mathbf{Y}}_{i,j} \right) \mid i = 1, \dots, F, j = 1, \dots, M \right\}. \quad (1)$$

Here, $\tilde{\mathbf{Y}}_{i,j}$ is the measured raw MET output sequence, and $\overset{\circ}{\mathbf{Y}}_{i,j}$ is the output sequence of an ideal linear system under the same frequency and magnitude conditions. The target sequence is generated by an ideal second-order linear reference system identified from the low-amplitude input regime. The preprocessing pipeline includes steady-state segment extraction, zero-mean correction, frequency and magnitude condition encoding, per-condition temporal training/validation splitting, input/target sequence normalization, and metric recomputation based on the same split, as shown in Fig. 2(b). The detailed protocol is provided in Supplementary Note 3.

The paired-sequence definition allows time-domain and frequency-domain errors to be computed from the same samples. Pointwise waveform error checks whether the compensated output follows the ideal reference sequence, while frequency-response recomputation examines whether the same model maintains stable natural frequency and sensitivity across magnitudes. The training/validation split is performed within each frequency–magnitude operating condition: the clipped steady-state sequence is divided into two temporal halves that are assigned to training and validation. Thus, every operating condition participates in both model training and evaluation, and the evaluation still covers the complete frequency–magnitude matrix.

3.3 Nonlinear Frequency Response Characteristics of MET

Fig. 2(c) shows the measured frequency response of the uncompensated MET under different input magnitudes. Across the 0.24–6.0 m/s² magnitude sweep, the fitted natural frequency spans 34.2–93.4 Hz, and the 100 Hz sensitivity spans 60.0–203.4 V · s/m. This reflects the significant amplitude-frequency coupling drift caused by the electrochemical-fluid system.

Although the full-range frequency response of the MET exhibits significant magnitude dependence, its measured frequency response at a fixed magnitude can still be approximated as a local linear second-order system. Based on this local linear characteristic, a three-branch parallel Wiener equivalent model can combine multiple linear dynamic kernels with static nonlinear mappings to represent the rightward shift of natural frequency and the sensitivity increase at 100 Hz as the input magnitude rises; it yields a center-frequency MAE of 1.87 Hz and a sensitivity MAE of 4.53 at 100 Hz. The detailed equivalent modeling process is provided in Supplementary Note 2.

4 Proposed Method

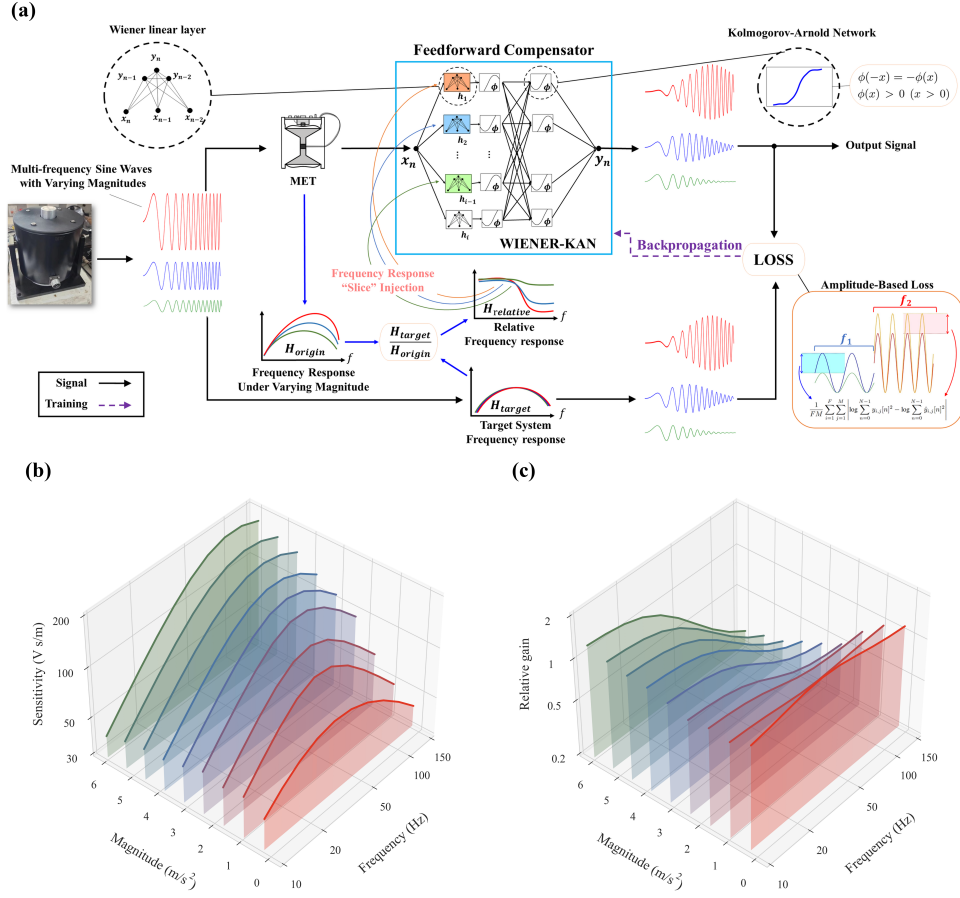


Fig. 3 Wiener-KAN compensation framework and magnitude-conditioned local transfer prior. (a) Schematic of the Wiener-KAN compensation framework. (b) Magnitude-conditioned local frequency responses identified from fixed-magnitude MET measurements. (c) Corresponding local relative transfer functions used to initialize the Wiener linear layer.

The proposed on-board real-time AI compensation system is illustrated in Fig. 3(a). Wiener-KAN casts the compensation problem into a neural Wiener system: a multi-kernel Wiener linear layer extracts frequency response features at different operating points to retain local physical priors, and a multi-layer, multi-input KAN network then learns the complex dynamic residuals and cross-branch crosstalk end to end, enabling global mapping reconstruction over a wide bandwidth and full dynamic range. Compared with traditional force feedback approaches, this feedforward structure requires no additional mechanical feedback unit, thereby avoiding closed-loop oscillation caused by nonlinearity in force feedback systems. The analytical difficulty of the parallel Wiener inverse model is given in Supplementary Note 4.

This structural choice follows the local-linearity observation described above. A single global filter has difficulty covering both the reference response at small magnitudes and the peak-frequency shift at large magnitudes, while a black-box time-series network lacks a stable pathway for injecting frequency response priors. Wiener-KAN cascades local linear dynamics with a trainable nonlinear mapping, allowing the frequency-domain prior to first project the input sequence into structured features and then allowing KAN to process the remaining magnitude-dependent mapping.

In the actual inference path, the input sequence first passes through the multi-kernel Wiener linear layer, producing a group of feature sequences associated with local dynamics at different magnitudes. KAN then performs multi-input nonlinear combination on these features, mapping local frequency response differences, magnitude-dependent calibration curves, and cross-branch residuals to the target linear output. This workflow preserves the physically interpretable structure of the Wiener model while avoiding manual derivation of a full-range analytical inverse model.

4.1 KAN-Based Nonlinear Mapping with Sign and Symmetry Constraints

To construct a compensator that integrates structured physical constraints with nonlinear modeling capability, this paper introduces trainable b-spline activation functions via KAN, using their piecewise local approximation property to directly learn the sensor calibration curve. Unlike MLPs that use fixed-shape activation functions for indirect fitting, the sign- and symmetry-constrained KAN nonlinear mapping adopted here optimizes the activation function shape through learnable b-spline coefficients, thereby enhancing the single-neuron modeling capacity for sensor nonlinearity [21].

$$\phi(x) = \text{sign}(x) \sum_{k=1}^K c_k B_k(|x|), \quad B_i(x) \geq 0 \quad \forall x \in \mathbb{R}, \quad c_k \geq 0. \quad (2)$$

Here, $B_k(x)$ denotes the b-spline basis function and c_k represents the trainable weights. Odd symmetry $\phi(-x) = -\phi(x)$ corresponds to the direction-consistent response of the MET sensor to positive and negative vibration inputs, while positivity $\phi(x) > 0$ ($x > 0$) corresponds to monotonic direction preservation within the positive input interval. The single-neuron example and the full expression of multi-layer KAN are provided in Supplementary Note 4.

These conditions align the function space of KAN with the directional behavior of the MET output. For positive and negative vibration inputs, the sensor readout polarity changes with the input direction; within one direction, the compensation mapping should maintain a stable directional relation as the magnitude increases. Embedding odd symmetry and positivity into the activation-function form reduces the possibility of sign reversal or locally implausible bending in magnitude intervals with sparse data.

4.2 Magnitude-Conditioned Local Transfer Functions for Wiener Layer Initialization

This paper implements the linear dynamic part of the Wiener structure as a trainable Wiener linear layer, initialized using magnitude-conditioned local transfer functions. Under local operating conditions, the MET can be approximated as a linear system. Therefore, for each fixed magnitude, this paper identifies its corresponding local linear frequency response from measurement data and calculates the corresponding local relative transfer function of the compensator at the same operating point according to the relative transfer function method for linear systems. By using these magnitude-varying local transfer functions as the initialization basis for the Wiener linear layer, this paper introduces signal-processing domain knowledge into the model to enhance its frequency-dependent compensation capability.

$$\text{Wiener-KAN}(x[n]) = \text{KAN}_{\text{con}} \circ \text{WienerLayer}(x[n]). \quad (3)$$

For different magnitude operating points, this paper constructs M mutually independent discrete linear kernels from the extracted difference equation parameters and applies these local linear filtering kernels in parallel to the same input signal, forming a single-input multiple-output (SIMO) multi-kernel Wiener linear feature extraction layer. During training, the weights of this linear layer are frozen, preserving the objective physical transfer prior of the sensor at different magnitudes. The local transfer function slices are shown in Fig. 3(b,c), and the discrete difference mapping is provided in Supplementary Note 4.

Freezing this linear layer has a specific role. The local transfer functions are derived from measured frequency response slices and represent the objective dynamics identified near each magnitude. Preserving these linear kernels during training prevents the network from arbitrarily changing the frequency response skeleton merely to reduce pointwise loss. The KAN back-end remains trainable to absorb dynamic residuals beyond the local linear approximation, cross-branch coupling, and the nonlinear part of the sensor calibration curve.

4.3 AFMAE Loss for Amplitude-Frequency Response

Conventional mean absolute error (MAE) and mean squared error (MSE) primarily measure pointwise error accumulation and tend to be overly sensitive to high-frequency random noise. For nonlinear system modeling, preserving the overall frequency response characteristics of the signal is more important. This paper therefore designs the amplitude-frequency mean absolute error (AFMAE) to quantify the difference between the compensated output and the ideal output in the frequency domain. AFMAE computes the logarithmic amplitude discrepancy from the output energy statistic under the same frequency and magnitude condition as

$$\mathcal{L}_{\text{AFMAE}} = \frac{1}{FM} \sum_{i=1}^F \sum_{j=1}^M \left| \log \sum_{n=0}^{N-1} y_{i,j}[n]^2 - \log \sum_{n=0}^{N-1} \hat{y}_{i,j}[n]^2 \right|. \quad (4)$$

Here, F is the number of frequency samples, and M is the number of magnitude samples. This method characterizes the amplitude-frequency response efficiently with a single energy statistic. In practice, a weighted combination of MAE and AFMAE yields superior results, and this paper introduces a weight hyperparameter β into the total loss function:

$$\mathcal{L}_{\text{total}} = (1 - \beta) \mathcal{L}_{\text{MAE}} + \beta \mathcal{L}_{\text{AFMAE}}. \quad (5)$$

The energy-amplitude derivation and logarithmic-form simplification of AFMAE are given in Supplementary Note 5.

The joint loss corresponds to two sources of error. MAE directly limits pointwise time-series deviation, helping preserve phase and local waveform consistency. AFMAE starts from the energy statistic under the same operating condition, making training sensitive to the discrepancy between the compensated amplitude-frequency response and the target response. When used together, the model must produce the correct time-domain waveform and maintain a stable amplitude response over the frequency–magnitude matrix.

This loss design is also consistent with the evaluation metrics used in this paper. Frequency drift and sensitivity drift are obtained by recomputing the compensated frequency response, while linearity error is derived from the in-band input-output amplitude relation. If the training objective focuses only on pointwise error, a model may obtain small local waveform deviation while retaining magnitude-dependent frequency-response differences. AFMAE brings this discrepancy into the optimization process, making the training objective closer to the final physical evaluation.

4.4 Precomputed LUT Inference for Real-Time Embedded Deployment

In typical seismic monitoring applications, continuous high-frequency sampling requires the single-step inference latency of the compensation algorithm to be smaller than the sampling period. The online inference cost of KAN mainly arises from evaluating the learnable activation function on each edge. To meet real-time requirements, this paper offline-samples this activation function into a precomputed lookup table after training, converting continuous spline evaluation into low-latency memory address mapping, table lookup, and optional first-order linear interpolation during online inference.

This deployment path differs fundamentally from the computational structure of an MLP. In an MLP layer, weights participate in online computation for every sample through matrix multiplication-addition, while edge weights in KAN are expressed by trainable functions. After training, the function shapes are precomputed into table values, and the online path mainly revolves around table address mapping, lookup, and optional interpolation. Consequently, the B-spline grid size and order primarily affect offline function expressiveness and table value generation, while the real-time feasibility of the exported implementation is directly evaluated by target-board throughput. The LUT grid, nearest-neighbor query, and linear-interpolation formulas are given in Supplementary Note 6.

In the embedded implementation, the LUT tables are written into the read-only C data region together with normalization constants, Wiener linear kernels, and output scaling parameters. During online operation, the linear recursion and KAN query path are executed point by point within a fixed sampling window, so the actual speed is affected by table access, function calls, and compiler optimization. This paper evaluates the exported implementation using QEMU-MAE, KEIL-MAE, KEIL speed, Flash, and RAM, allowing numerical consistency and real-time feasibility to be verified by the same C kernel.

5 Experiments and Results

In all experiments, models were trained with TensorFlow and the Adam optimizer, with batch size set to 260000 and total training epochs set to 1000. The model adopted fixed learning rate, the learning rate $\alpha(k)$ was set to 0.0007, and β in the loss function was set to 0.2.

Three physical metrics are used to evaluate compensation quality, while deployment feasibility is assessed via on-board numerical consistency, measured throughput, and memory footprint. Frequency drift is defined as the maximum deviation of the fitted natural frequency from its median across different excitation levels within 10–128 Hz. Sensitivity drift is defined as the maximum deviation of the interpolated sensitivity at 100 Hz from its median across different excitation levels, with units of $V \cdot s/m$. Linearity error is defined as the mean percentage of $1 - R^2$ in the input-output amplitude fitting below 128 Hz. KEIL speed is the measured on-board throughput, while QEMU-MAE and KEIL-MAE verify numerical consistency between the exported C implementation and the TensorFlow reference output.

5.1 Wiener-KAN Compensation

This section focuses on evaluating the proposed Wiener-KAN compensation model. The uncompensated MET response exhibits a frequency drift of 34.50 Hz, a sensitivity drift of $84.56 V \cdot s/m$, and a linearity error of 2.004% over the same frequency–magnitude matrix; Wiener-KAN reduces these three metrics to 2.34 Hz, $9.09 V \cdot s/m$, and 0.511%, respectively.

The three metrics cover the main observable manifestations of large-magnitude MET nonlinearity. Frequency drift reflects the migration of the resonance position with input magnitude, sensitivity drift reflects the magnitude dependence of passband gain at a fixed frequency, and linearity error reflects the deviation of the input-output amplitude relation from the linear reference within the in-band range. The simultaneous reduction of all three metrics indicates that Wiener-KAN improves the response across multiple frequency points and waveform segments over the frequency–magnitude matrix.

Fig. 4(a) shows the amplitude-frequency response curves before and after compensation, where the response envelopes under different excitation levels are significantly narrowed. Fig. 4(b)–(c) illustrate the suppression of amplitude-dependent frequency response drift by Wiener-KAN. After compensation, the natural frequency trajectories and sensitivity trajectories at 100 Hz converge toward the low-amplitude reference response. Fig. 4(d) presents representative time-domain waveforms, where predicted waveforms match the ideal reference output in both phase and amplitude. Fig. 4(e) shows the overall reduction in frequency drift and sensitivity drift.

From the curve shape, the compensated response preserves the main bandpass shape of the low-amplitude reference system while reducing the peak-position and gain differences

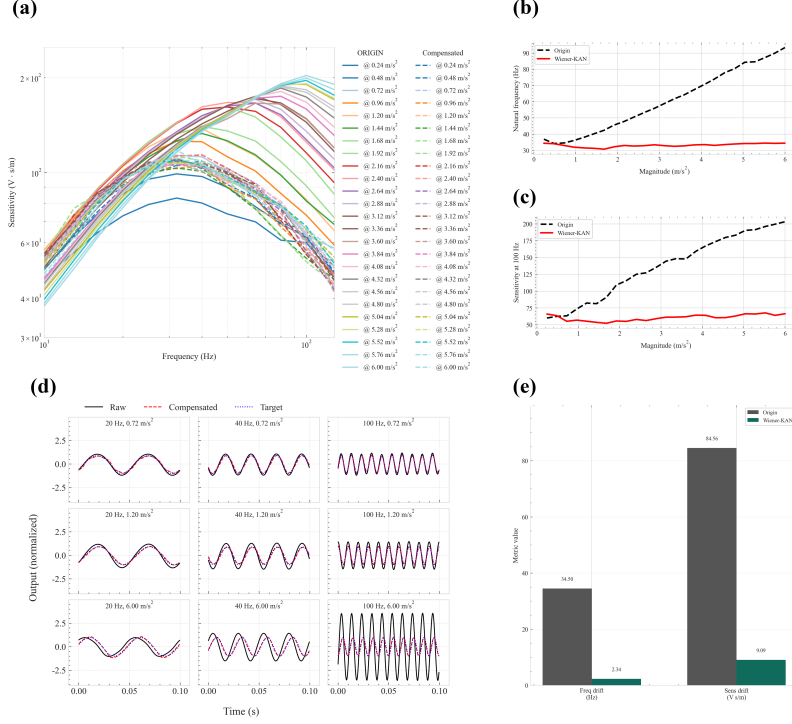


Fig. 4 Effectiveness of the proposed Wiener-KAN compensation model in reducing amplitude-dependent frequency-response drift. (a) Frequency-response curves before and after compensation under representative magnitude conditions. (b) Natural-frequency trajectory versus magnitude before and after compensation. (c) Sensitivity trajectory at 100 Hz before and after compensation. (d) Representative time-domain outputs under selected frequency and magnitude conditions. (e) Frequency-drift and sensitivity-drift values before and after compensation.

induced by large-magnitude inputs. The time-domain waveform alignment further indicates that the frequency-domain convergence is achieved while preserving phase consistency. This result is consistent with the method design: the Wiener linear layer introduces the local frequency response prior, and the KAN mapping corrects the remaining magnitude-dependent nonlinearity.

5.2 Comprehensive Comparison with Baseline Models

To evaluate the comprehensive performance of Wiener-KAN, this paper compares it with representative nonlinear compensation methods, including the conventional Wiener model and time-series learning baselines such as 1DCNN, TCN, RNN, GRU, LSTM, and LSTM-Transformer. All baselines adopt the same data split, training epochs, and fixed learning rate strategy as the proposed Wiener-KAN, with width, depth, convolution kernel, and recurrent unit counts selected under comparable parameter budgets or resource constraints to maintain scale comparability in the baseline comparison.

Tab. 1 compares Wiener-KAN with typical time-series baselines under a unified MET compensation protocol. Wiener-KAN simultaneously achieves the lowest frequency drift,

Table 1 Main comparison under the physical-calibration and on-board efficiency protocol. Lower values are better for frequency drift, sensitivity drift, linearity error, and RAM; higher values are better for KEIL speed.

Method	Freq drift (Hz)	Sens drift (V · s/m)	Linearity (%)	KEIL speed (points/s)	RAM (KB)
Wiener	34.50	84.56	2.004	–	–
1DCNN	3.72	16.24	0.920	2009.9	50.9
TCN	2.65	16.08	0.517	935.7	106.7
RNN	6.68	23.75	0.726	3652.9	58.7
GRU	7.40	27.15	0.726	1660.1	58.7
LSTM	4.50	13.80	0.512	1426.3	58.7
LSTM-Trans.	4.72	18.38	0.567	1461.5	83.7
Wiener-KAN	2.34	9.09	0.511	6986.7	5.6

lowest sensitivity drift, and highest on-board throughput; TCN approaches Wiener-KAN in linearity error, but its KEIL speed is lower than that of the proposed model; RNN offers high on-board throughput, yet its suppression of physical compensation metrics is limited.

This comparison shows that physical compensation accuracy and on-board efficiency must be examined together. Convolutional and recurrent models can learn temporal correlations, but their frequency response skeletons are mainly formed by data-driven optimization and are therefore sensitive to the training-sample distribution when maintaining frequency response across magnitudes. Wiener-KAN reduces the frequency-domain learning burden through explicit local transfer-function initialization and reduces online computation through LUT inference, maintaining advantages in both physical metrics and target-board throughput.

Fig. 5(a)–(c) show the natural frequency, sensitivity at 100 Hz, and in-band linearity error ranges of representative models. Wiener-KAN exhibits the most convergent distributions in natural frequency and sensitivity. Fig. 5(d)–(f) present the on-board measured throughput, a five-metric radar chart, and the training convergence curves of the baseline comparison, respectively. The radar chart normalizes frequency drift, sensitivity drift, linearity error, KEIL speed, and KEIL RAM onto common coordinates, showing that Wiener-KAN maintains a more balanced metric profile between physical compensation and embedded efficiency.

The training convergence curves support this conclusion. Since all models use the same data split and training setting, the loss trajectories mainly reflect how each structure adapts to the same compensation task. The convergence result of Wiener-KAN is consistent with the physical metrics, indicating that the local frequency response prior, sign- and symmetry-conditioned KAN, and joint loss act together on the final compensation quality.

From the deployment perspective, the horizontal comparison also shows that model accuracy must be evaluated together with the exported implementation. Some time-series models provide a degree of compensation in the physical metrics, but their online paths depend on recurrent state updates or convolutional computation, so target-board throughput and memory footprint constrain their use in continuous sampling. The LUT path of Wiener-KAN moves the main nonlinear computation to offline table generation, allowing the same model to combine frequency-response compensation with low-latency execution.

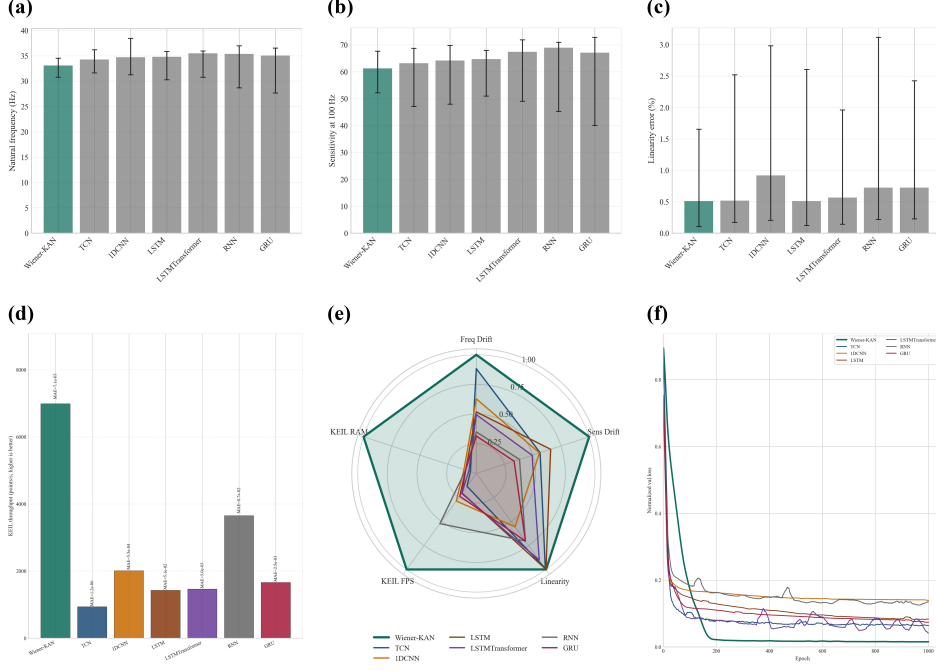


Fig. 5 Main benchmark summary. (a) Natural-frequency range across magnitudes. (b) Sensitivity range at 100 Hz across magnitudes. (c) In-band linearity-error range across frequency points. (d) STM32F405 measured KEIL throughput with exported-kernel error annotations. (e) Five-metric radar comparison across physical accuracy and on-board efficiency metrics. (f) Main-benchmark validation-loss trajectories. Error bars in (a)–(c) indicate the corresponding min–max ranges.

5.3 Ablation Studies of Wiener-KAN Components

The ablation study evaluates the training objective, structural variants, sign and symmetry conditions, Wiener front-end, and hyperparameter settings of Wiener-KAN. Tab. 2 summarizes 10 numerical comparisons covering structural variants, sign and symmetry conditions, the Wiener front-end, and the loss function.

The structural ablation indicates that CNN-KAN and Wiener-MLP correspond to the linear dynamic skeleton and nonlinear compensation body of Wiener-KAN, respectively. CNN-KAN approaches the baseline configuration in frequency drift and sensitivity drift, yet exhibits a marked increase in linearity error, suggesting that the Wiener front-end initialized with local frequency response is more conducive to preserving the stable shape of the in-band compensation curve. Wiener-MLP lags behind Wiener-KAN on all three physical metrics, indicating that after sharing the Wiener front-end, the physics-constrained KAN mapping still undertakes the primary expressive capability for amplitude-dependent nonlinear compensation.

The constraint ablation study examines odd symmetry and positivity separately. The *Odd/No positive basis* variant in Tab. 2 retains odd symmetry but removes the direction-preserving constraint on the positive input interval, increasing frequency drift, sensitivity drift, and linearity error simultaneously; this indicates that the positivity constraint provides Wiener-KAN

Table 2 Numerical overview of Wiener-KAN ablation studies. Each row reports the evaluated physical-metric triplet for one design contrast.

Group	Variant	Controlled contrast	Freq drift (Hz)	Sens drift (%)	Linearity (%)
Baseline	Canonical	B-spline KAN; odd + positive; fixed system prior; MAE+AFMAE	2.34	9.09	0.511
Struct	CNN-KAN	Conv front replaces Wiener front	2.43	9.22	0.883
	Wiener-MLP	MLP replaces KAN mapping	5.82	27.48	0.782
Constr.	Positive	No odd symmetry	10.91	10.61	2.109
	Odd	No positive basis	58.03	50.87	38.586
Wiener front	Random/fixed	Random init., frozen	5.37	22.28	1.308
	Sys/trainable	Measured init., trainable	94.87	64.68	7.478
	Rand/trainable	Random init., trainable	102.57	16.65	5.144
Loss	MAE	Time-domain pointwise loss	13.00	20.28	0.793
	AFMAE	Amplitude-frequency loss	4.00	21.06	1.068

with a stable functional shape prior. The Wiener front group further shows that the frozen system prior setting yields the best frequency drift, sensitivity drift, and linearity error, while the trainable IIR front-end causes a marked increase in frequency drift.

The loss-function ablation shows that MAE+AFMAE yields the most balanced results across frequency drift, sensitivity drift, and linearity error, indicating that pointwise time-domain error and amplitude-frequency response error must jointly participate in the training objective. With MAE alone, the pointwise error objective imposes weak constraints on amplitude-related frequency drift. With AFMAE alone, the amplitude-frequency statistical constraint improves frequency drift, but sensitivity drift and linearity error remain higher than those of the joint objective. Detailed structural, constraint, Wiener-front-end, hyperparameter, and loss-function ablations are provided in Supplementary Note 7.

These ablation results jointly define the role of each component. The Wiener front-end mainly determines the frequency-domain dynamic skeleton, the KAN mapping mainly determines the magnitude-dependent nonlinear expression, the sign and symmetry conditions stabilize the function shape, and MAE+AFMAE brings both time-domain waveform error and amplitude-frequency error into training. When any component is replaced or removed, at least one physical metric increases, showing that the final performance is jointly determined by the structural prior, functional conditions, and training objective.

On-board inference testing verifies the complete pipeline of Wiener-KAN from the trained model to the embedded C implementation. After training, the weights, normalization constants, and LUT tables are exported as C code. The same C implementation is first run in QEMU and compared with the TensorFlow reference output, then flashed onto the STM32F405 target board. Throughput is collected via a serial benchmark loop, and Flash and RAM usage are read from the Keil build results. The primary Wiener-KAN implementation achieves 6986.7 points/s with 5.6 KB RAM. The detailed verification flow, LUT variants, and quantization-point sweep are provided in Supplementary Note 6.

The on-board results provide direct evidence for real-time feedforward compensation. The QEMU-to-TensorFlow consistency check excludes numerical deviation in the exported kernel, while the target-board serial benchmark measures the actual runtime speed including memory access and function-call overhead. Because the LUT path converts spline-function evaluation into table access, the primary implementation achieves high throughput with a small RAM footprint, satisfying the pointwise compensation target in a continuous sampling chain.

6 Discussion

The validation in this paper is based on one MET sensor sample, a controlled sinusoidal excitation matrix, and an ambient temperature of 25°C. This setup provides a reproducible frequency–magnitude response matrix for quantifying magnitude-dependent drift; additional temperature conditions, more sensor samples, and composite waveforms resembling field seismic records require independent measurement data. The embedded results are verified through QEMU numerical checks and STM32F405 on-board measurements of the exported implementation. KEIL speed and KEIL RAM Usage describe the implementation results under the current compiler, MCU, numerical precision, and window length; QEMU-MAE and KEIL-MAE are used to check numerical consistency between the exported C implementation and the TensorFlow reference output. Other MCU families, fixed-point implementations, multi-channel acquisition, and different compilation settings will alter the measured throughput and memory footprint.

These applicability conditions preserve the core conclusion on method feasibility and delimit the scope of the results. The reported frequency drift, sensitivity drift, and linearity error should be interpreted as compensation results under the current sample, frequency–magnitude matrix, and target reference system; the on-board speed and memory footprint should be interpreted as implementation results under the current STM32F405, floating-point realization, and exported kernel. When transferring the workflow to other MET structures, the local frequency responses should be remeasured, the target reference system rebuilt, and the same metrics recomputed.

6.1 Code Availability

The code used for model training, evaluation, embedded export, data partitioning, and metric calculation is publicly available at <https://github.com/pikastech/wiener-kan>.

6.2 Data Availability

Existing public sensor drift datasets mainly cover other sensor types, such as gas sensor arrays [29]. To our knowledge, no public resource provides synchronized MET output, reference excitation, frequency, magnitude, and target calibration curves for amplitude-dependent time-domain compensation. The dataset used in this paper is a self-built frequency–magnitude response matrix for MET drift compensation. The processed response matrix and representative waveforms are publicly available at <https://github.com/pikastech/wiener-kan>.

7 Conclusion

This work constructs the Wiener-KAN nonlinear compensation system by combining the linear-nonlinear decomposition of the Wiener structure, frequency response information, and Kolmogorov-Arnold networks. This paper constructs and releases a three-dimensional frequency response dataset capturing amplitude-frequency coupling effects, maps relative frequency response slices into the initialization weights of the Wiener linear layer, uses trainable spline activation functions with sign and symmetry constraints to directly learn the sensor calibration curve, and constrains amplitude-frequency discrepancy through AFMAE. Experimental results show that Wiener-KAN improves the input-output curve of electrochemical seismometers and suppresses frequency drift, indicating that feedforward compensation can improve the large-magnitude response quality of MET sensors while preserving the sensor hardware structure. The inference mechanism based on a precomputed lookup table further demonstrates the feasibility of real-time on-board computation in seismic instruments.

At the full-pipeline level, this paper connects amplitude-frequency coupling tests, local frequency response priors, sign- and symmetry-conditioned KAN nonlinear mapping, the AFMAE training objective, and LUT-based on-board inference into a closed workflow. The workflow preserves the interpretable part of the MET small-signal frequency response model while assigning the analytically difficult magnitude-dependent nonlinearity to trainable functions. Wiener-KAN therefore provides a reproducible path for real-time large-magnitude compensation of electrochemical seismometers without adding force-feedback hardware.

Supplementary Information

Supplementary information accompanies the manuscript on the Microsystems & Nanoengineering website (<http://www.nature.com/micronano>).

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Author Contributions

A.L. led the methodology development, software implementation, data analysis, visualization, and writing of the original draft. H.Y. contributed to the methodology, provided experimental resources and conditions, supervised the project, and reviewed and edited the manuscript. H.Z. contributed to supervision, experimental resources, and manuscript review and editing. X.D. contributed to resources, validation, and manuscript review and editing. F.Z. and L.Z. contributed to software implementation and validation. C.L. and R.L. contributed to data collection, data curation, and validation. All authors reviewed and approved the final manuscript.

Competing Interests

The authors declare no competing interests.

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